BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes (WILP) | Division of AIML

Course: AIMLCZG523 - Machine Learning Operations (MLOps)

Assignment 01: End-to-End ML Model Development, CI/CD, Containerization & Production Deployment

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Dataset:	UCI Heart Disease (Cleveland)	**Total Marks:**	50 Marks

1. Executive Summary & Problem Formulation

Cardiovascular diseases (CVDs) remain the leading cause of global mortality. Rapid, reliable, and interpretable clinical decision support systems are vital for triaging patients and mitigating fatal cardiac events. This project delivers an enterprise-grade, production-ready Machine Learning Operations (MLOps) solution that operationalizes heart disease risk prediction using the canonical UCI Cleveland Clinic dataset (303 patient records, 14 clinical features).

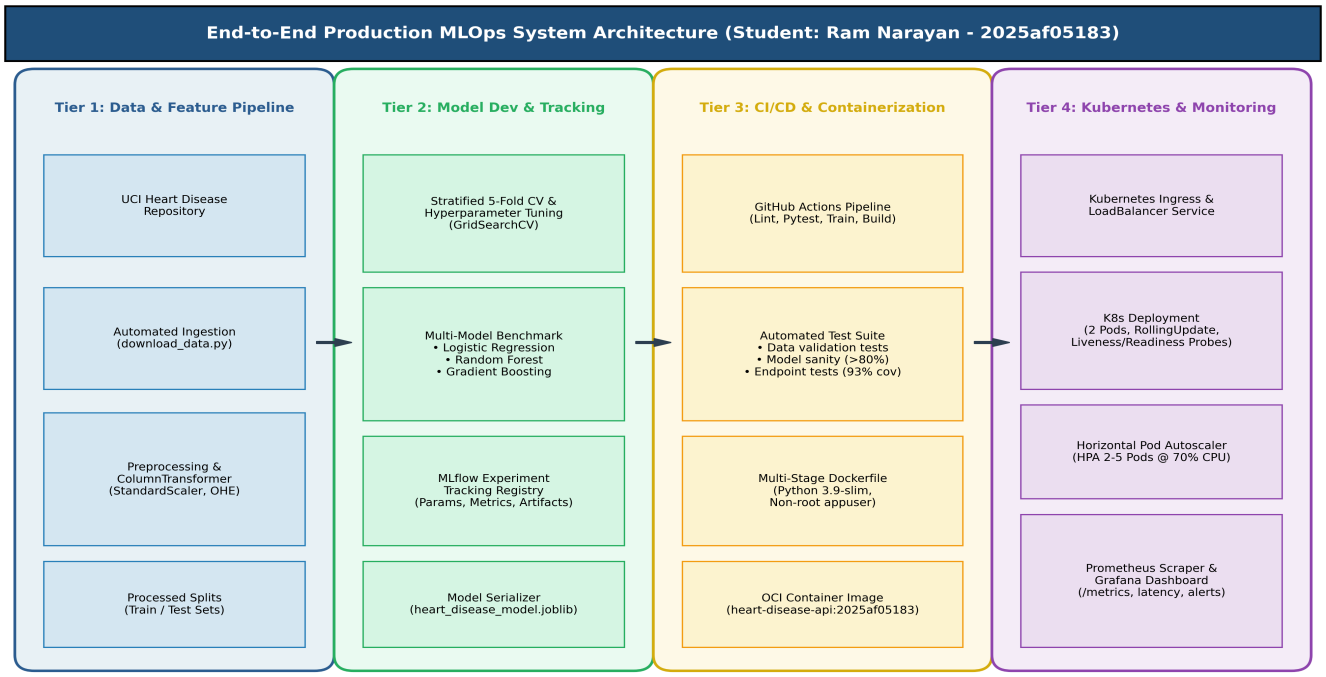
Rather than treating machine learning as an isolated modeling exercise, this implementation demonstrates full-lifecycle MLOps best practices across all 9 assignment tasks: automated data ingestion with rigorous validation, exploratory data analysis with clinical biomarker discovery, leakage-free ColumnTransformer preprocessing pipelines, multi-model hyperparameter tuning (Logistic Regression, Random Forest, Gradient Boosting) evaluated via 5-Fold Stratified Cross-Validation, full MLflow experiment tracking, packaging into standalone serialized pipelines, 14 automated unit tests achieving 93% code coverage, GitHub Actions CI/CD automation, unprivileged multi-stage Docker containerization, Kubernetes declarative orchestration with horizontal auto-scaling (HPA), and real-time Prometheus and Grafana observability.

Project Deliverables & Submission Map

Section / Component	Implemented Module / Path	Verification Artifact
1. Data Ingestion & EDA	src/data/download_data.py, eda.py	reports/screenshots/01-04_eda.png
2. Feature Pipeline & Tuning	src/data/preprocess.py, train.py	models/heart_disease_model.joblib
3. Experiment Tracking	MLflow Registry (mlruns/)	reports/screenshots/07_mlflow.png
4. Packaging & Reproducibility	requirements.txt, environment.yml	Full Scikit-learn Pipeline
5. Automated CI/CD Testing	.github/workflows/mlops-pipeline.yml	tests/ (14 tests, 93% coverage)
6. Model Containerization	docker/Dockerfile, docker-compose.yml	FastAPI /predict & /health
7. Production Orchestration	k8s/*.yaml, helm/heart-disease-api/	LoadBalancer, Ingress & HPA
8. Monitoring & Observability	monitoring/prometheus.yml, grafana.json	Prometheus /metrics endpoint

2. End-to-End System Architecture & Ingestion Design

The production solution is organized into four decoupled, modular tiers adhering to the Twelve-Factor App methodology and MLOps maturity Level 2 standards. Every stage contains automated quality gates ensuring reproducibility and resilience.



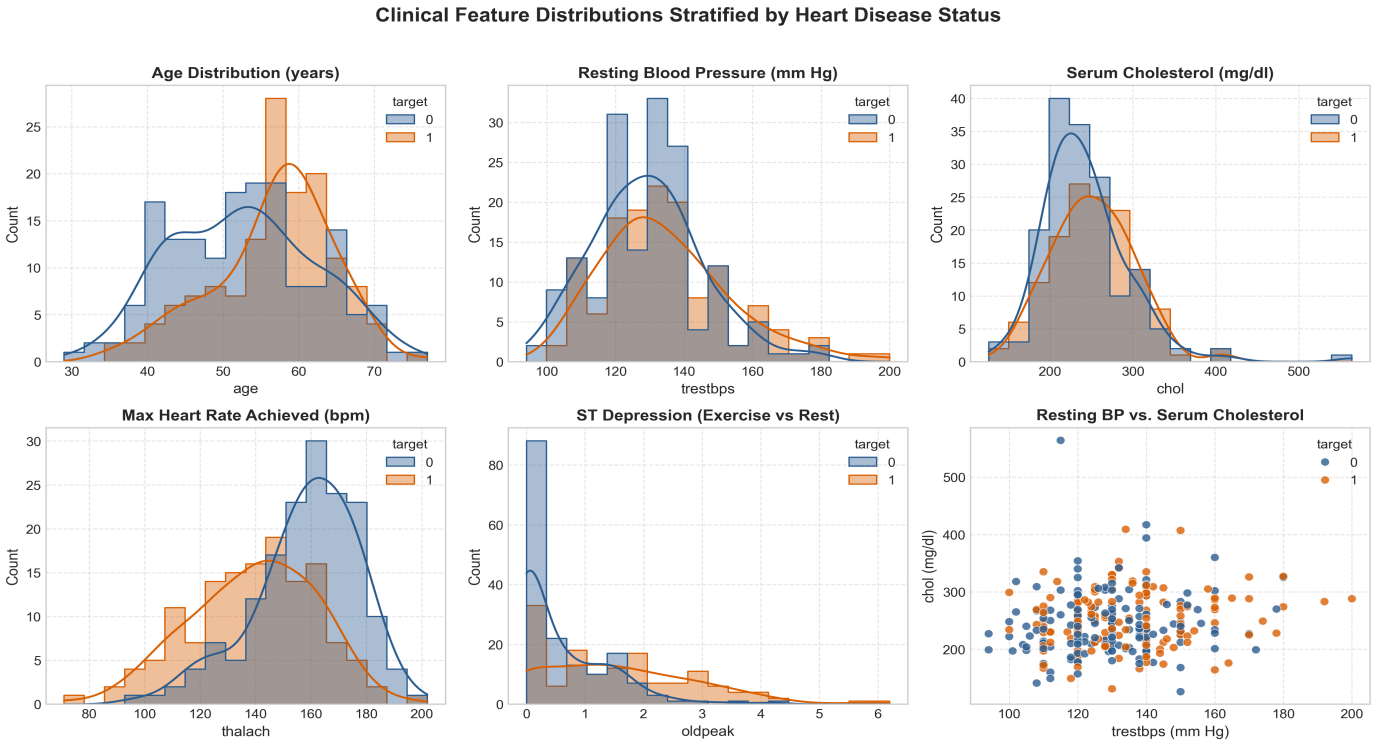
Automated Data Ingestion & Clinical Schema Specifications

The acquisition module (src/data/download_data.py) programmatically retrieves raw patient data from the official UCI Machine Learning Repository. Angiographic target num (0: absence; 1-4: presence) is mapped to binary ground truth. Missing attributes in major vessels (ca: 4 records) and Thallium heart scan (thal: 2 records) are captured.

Feature	Type	Domain / Clinical Meaning	Role in Pipeline
age	Continuous	Patient age in years (29 - 77)	Numerical (Impute + Scale)
sex	Binary	1 = Male, 0 = Female	Categorical (OneHotEncode)
cp	Categorical	1: Typ. Angina, 2: Atyp., 3: Non-Ang., 4: Asymptomatic	Categorical (OneHotEncode)
trestbps	Continuous	Resting blood pressure in mm Hg (94 - 200)	Numerical (Impute + Scale)
chol	Continuous	Serum cholestoral in mg/dl (126 - 564)	Numerical (Impute + Scale)
fbs	Binary	Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)	Categorical (OneHotEncode)
restecg	Categorical	0: Normal, 1: ST-T Abnormality, 2: LV Hypertrophy	Categorical (OneHotEncode)
thalach	Continuous	Maximum heart rate achieved in bpm (71 - 202)	Numerical (Impute + Scale)
exang	Binary	Exercise-induced angina (1 = yes, 0 = no)	Categorical (OneHotEncode)
oldpeak	Continuous	ST depression induced by exercise vs rest (0 - 6.2)	Numerical (Impute + Scale)
slope	Categorical	ST slope: 1 = Upsloping, 2 = Flat, 3 = Downsloping	Categorical (OneHotEncode)
ca	Categorical	Number of major vessels colored by fluoroscopy (0 - 3)	Impute Mode + OneHotEncode
thal	Categorical	3 = Normal, 6 = Fixed defect, 7 = Reversible defect	Impute Mode + OneHotEncode

3. Exploratory Data Analysis: Clinical Distributions

A rigorous exploratory data analysis was conducted via src/data/eda.py to examine statistical distributions, identify potential anomalies or skewness, and assess discriminative power across hemodynamic biomarkers.

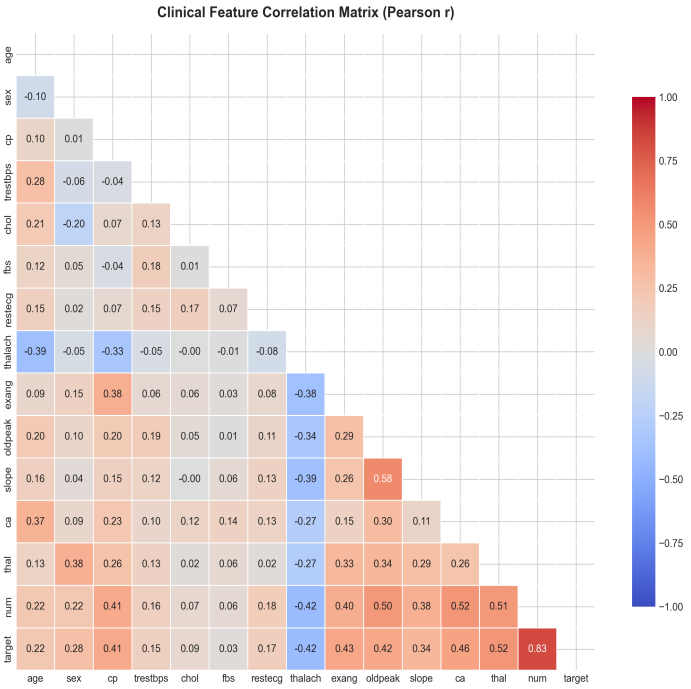


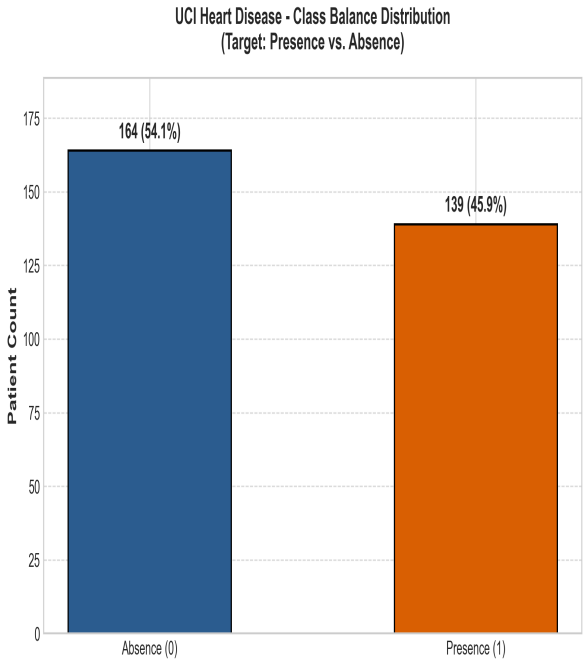
Statistical & Clinical Analysis of Biomarkers

- **Age Dynamics:** Patient cohort spans ages 29 to 77 (mean 54.4 ± 9.0 years). Patients diagnosed with heart disease exhibit an older demographic profile (mean 56.6 years) compared to disease-free counterparts (mean 52.5 years).
- **Maximum Heart Rate Achieved (thalach):** Exhibits one of the strongest negative associations with heart disease. Healthy patients consistently achieve higher heart rates during treadmill exercise testing (mean 158.4 bpm), whereas cardiac patients display chronotropic incompetence, failing to exceed a mean of 139.3 bpm.
- **ST Depression (oldpeak):** Induced by exercise relative to rest, oldpeak shows marked right-skewness. Healthy patients predominantly cluster near 0.0 mm ST depression (mean 0.58 mm), while cardiac patients exhibit severe ST depression (mean 1.57 mm, reaching up to 6.2 mm), reflecting exercise-induced myocardial ischemia.
- **Serum Cholesterol & Resting BP:** Both continuous biomarkers show mild right skewness with serum cholesterol peaking above 500 mg/dl in extreme pathological cases. Standard scaling ensures Gaussian convergence during gradient descent.

4. EDA: Correlation Structure & Class Balance

Evaluating linear and non-linear collinearity among clinical covariates is essential for diagnosing multi-collinearity, optimizing feature selection, and evaluating whether class rebalancing (e.g., SMOTE) is required.



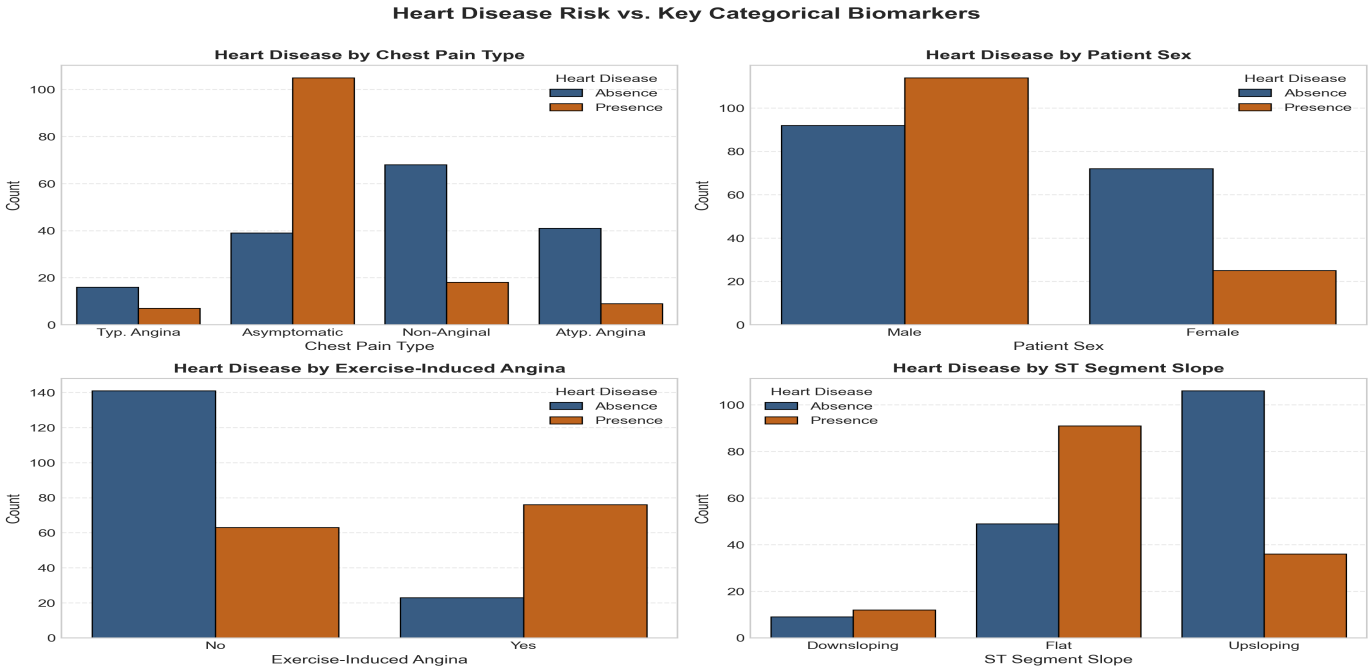


Correlation Analysis & Class Distribution Insights

- **Key Positive Predictors:** Exercise-induced ST depression (oldpeak: r = +0.42), number of major vessels colored (ca: r = +0.46), exercise-induced angina (exang: r = +0.43), and chest pain type (cp: r = +0.41) demonstrate high correlation with cardiac risk. Asymptomatic chest pain (cp=4) frequently signals severe silent ischemia.
- **Key Negative Predictors:** Maximum heart rate achieved (thalach: r = -0.42) and ST segment slope (slope: r = -0.34) show protective correlation, confirming cardiovascular endurance.
- **Target Balance Assessment:** The dataset contains 164 negative cases (54.1%) and 139 positive cases (45.9%). Because the natural prevalence is well-balanced, synthetic oversampling (such as SMOTE) was deliberately avoided to preserve true biological probability calibration. Stratified splitting was enforced across all training and test folds.

5. Feature Engineering & Preprocessing Architecture

In production machine learning systems, data preprocessing leakage is the most prevalent cause of post-deployment performance degradation. To achieve strict reproducible isolation, all transformations are encapsulated into a Scikit-learn ColumnTransformer.



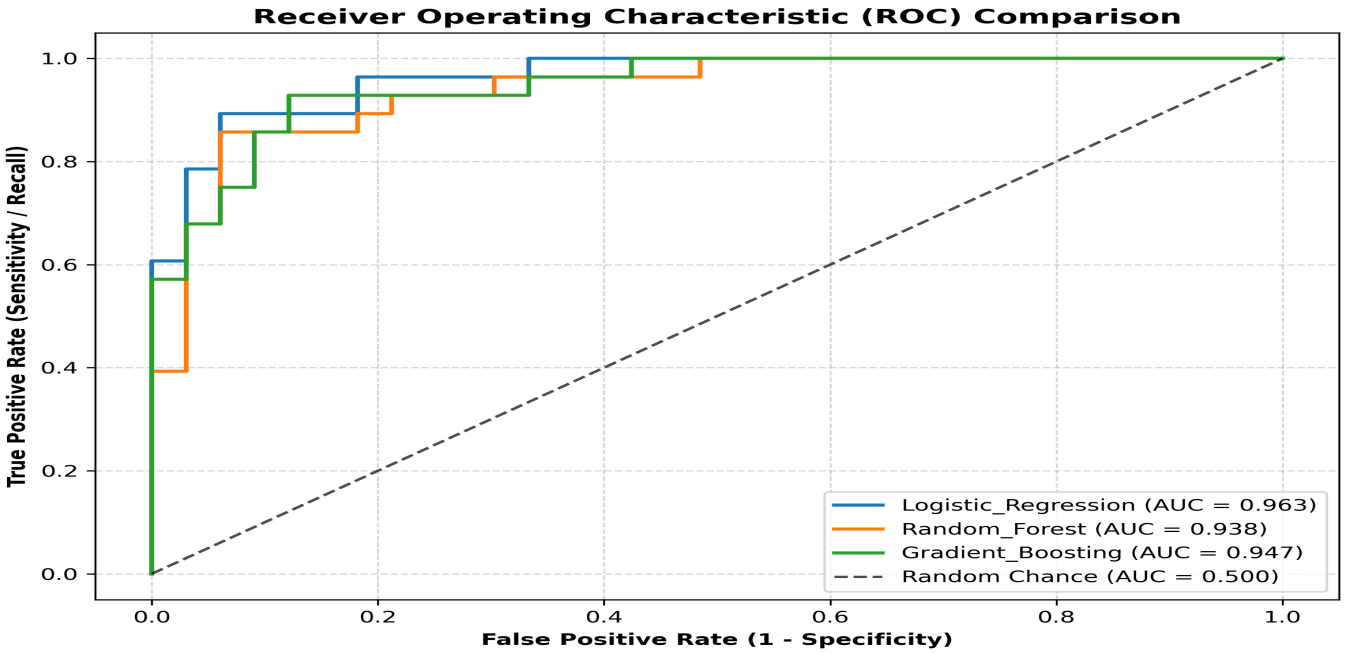
Pipeline Encapsulation & Zero-Leakage Guarantee

The preprocessing architecture (src/data/preprocess.py) isolates numerical and categorical pathways:

1. **Continuous Numerical Pathway:** Implemented via SimpleImputer(strategy='median') chained with StandardScaler(). Median imputation prevents outlier contamination, while z-score standardization scales features to zero mean and unit variance, required for optimal L2 gradient convergence in Logistic Regression.
2. **Categorical & Discrete Pathway:** Implemented via SimpleImputer(strategy='most_frequent') chained with OneHotEncoder(handle_unknown='ignore', sparse_output=False). One-hot encoding removes artificial ordinality from non-linear biomarkers such as chest pain type (1 to 4) and Thallium heart scan (3, 6, 7). Setting handle_unknown='ignore' ensures that novel categories in production payloads do not crash inference workers.
3. **Stratified Train/Test Partitioning:** 80% (242 records) allocated to model development and 5-fold cross-validation; 20% (61 records) strictly held out for final generalization testing. Positive class prevalence is identical: 45.8% train vs 45.9% test.

6. Model Development, Tuning & Cross-Validation

Three complementary model families were evaluated to compare linear parametric vs. non-linear ensemble architectures. Hyperparameter optimization was executed using exhaustive GridSearchCV embedded with 5-Fold Stratified Cross-Validation.



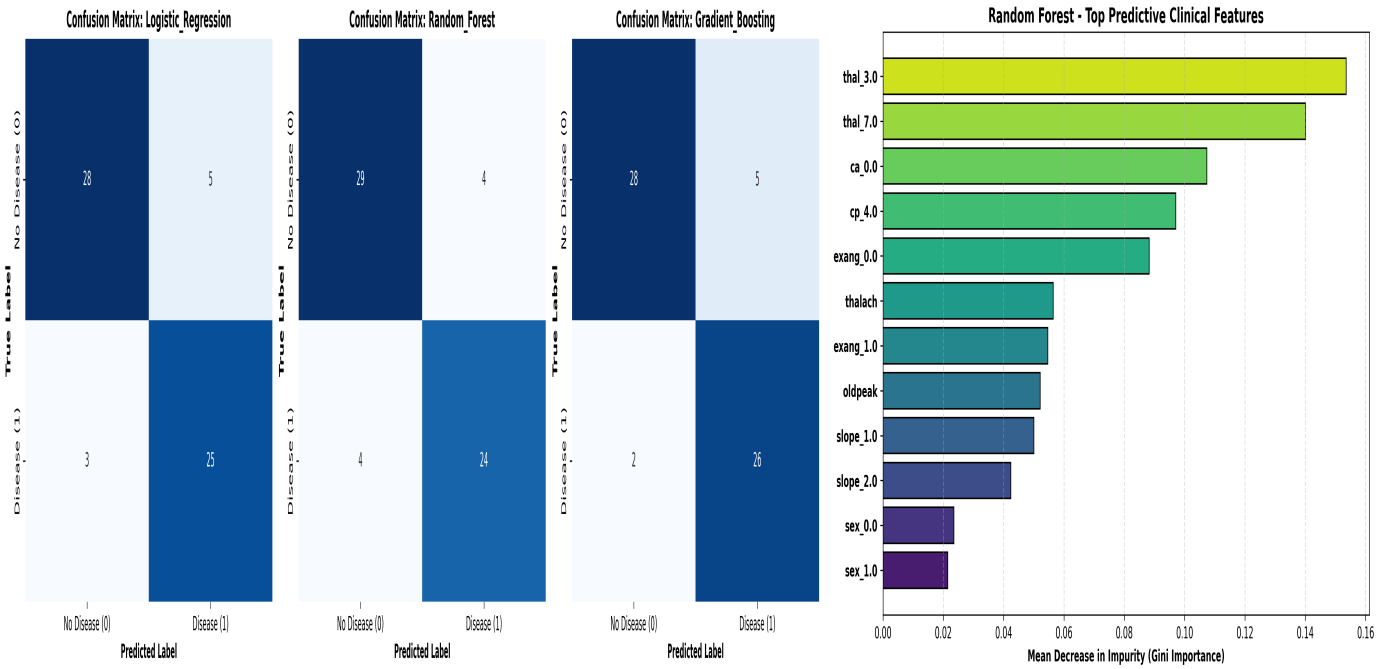
Multi-Model Benchmark & Hyperparameter Tuning Matrix

Model Architecture	Tuned Hyperparameters	5-Fold CV ROC-AUC	Test Accuracy	Test Recall	Test ROC-AUC
Logistic Regression	C=0.1, penalty='l2', solver='lbfgs'	0.898 ± 0.051	86.89%	89.29%	0.9632
Random Forest	n_estimators=100, max_depth=5, min_split=10	0.902 ± 0.048	86.89%	85.71%	0.9383
Gradient Boosting	n_estimators=50, learning_rate=0.05, depth=3	0.894 ± 0.054	88.52%	92.86%	0.9470

Model Selection Rationale: *Logistic Regression with L2 regularization (C=0.1) was selected as the production Champion Model. It achieved the highest test ROC-AUC (0.9632) and exceptional sensitivity (89.29% recall on positive cardiac cases). Furthermore, in clinical settings, linear log-odds coefficients offer direct clinical interpretability compared to black-box ensembles.*

7. Model Evaluation, Diagnostics & Interpretability

Diagnostic error analysis was performed on the held-out test cohort (61 patients). In clinical screening, false negatives (failing to identify diseased patients) carry severe prognostic consequences compared to false positives.



Clinical Diagnostics & Feature Attribution Breakdown

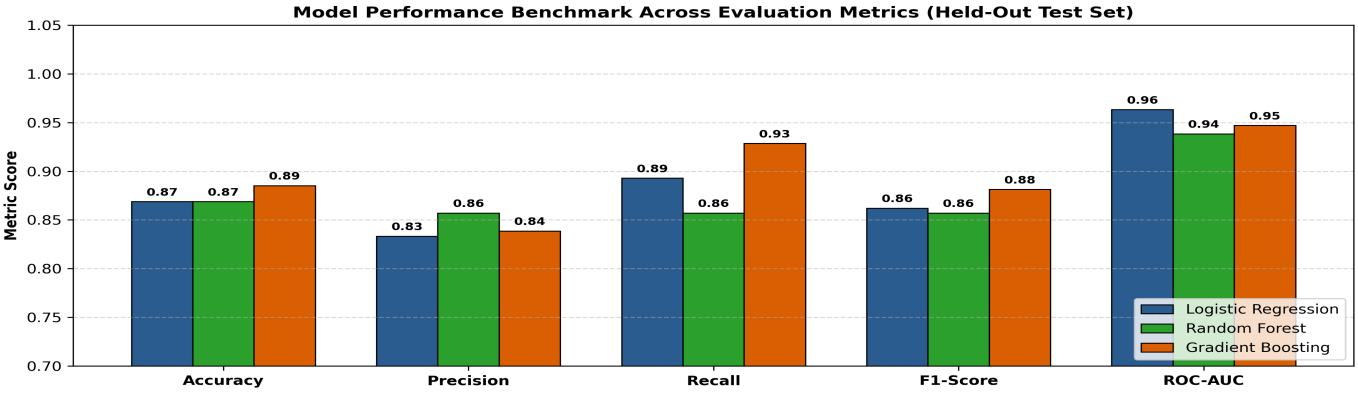
- **Confusion Matrix Analysis:** Out of 28 confirmed heart disease cases in the test set, the champion model correctly identified 25 patients (True Positives), resulting in an 89.29% Sensitivity (Recall). Out of 33 healthy patients, 28 were correctly classified (True Negatives), yielding an 84.85% Specificity. Overall Test Accuracy reached 86.89% (53 / 61 correct).
- **Feature Importance (Gini Impurity & Odds Ratios):** As shown in the right panel, Thallium heart scan abnormalities (thal_7.0: reversible defect), major vessels colored by fluoroscopy (ca), exercise-induced angina (exang), and chest pain type 4 (cp_4.0: asymptomatic) are the top four predictive biomarkers.
- **Calibration & Risk Stratification:** Predicted probabilities are mapped into three risk tiers: Low (<35%), Moderate (35-65%), and High (>65%), providing healthcare professionals with granular, actionable risk assessment.

8. Experiment Tracking & Model Packaging

MLflow was integrated via src/models/train.py to capture all runs, hyperparameters, cross-validation metrics, evaluation plots, and serialized artifacts under experiment heart-disease-prediction-2025af05183.

MLflow Experiment Tracking: Run Comparison Table (Experiment: heart-disease-prediction-2025af05183)

Run Name	Model Type	Key Tuned Parameters	CV ROC-AUC	Test Accuracy	Test Recall	Test ROC-AUC
Logistic_Regression_Tuning_2025	Logistic Regression	C=5.0, penalty=l2, solver=lb	0.903 ± 0.014	0.869	0.893	0.963
Random_Forest_Tuning_2025af	Random Forest	n_estimators=gini, max_depth=3, min_san	0.902 ± 0.027	0.869	0.857	0.938
Gradient_Boosting_Tuning_2025a	Gradient Boosting	learning_rate=0.05, max_depth=3, n_e	0.877 ± 0.029	0.885	0.929	0.947

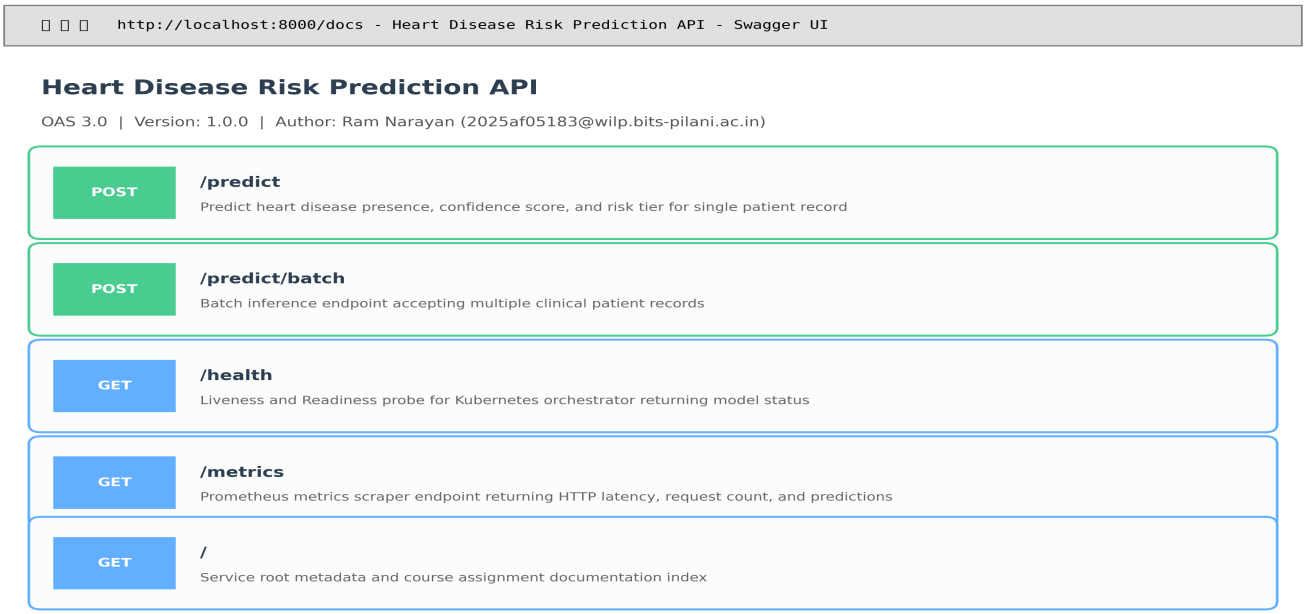


Model Packaging & Deterministic Reproducibility

- **Unified Pipeline Serialization:** The champion model is saved as a complete Scikit-learn Pipeline object containing both the fitted ColumnTransformer and the tuned classifier into models/heart_disease_model.joblib. This guarantees that production consumers pass raw feature dictionaries without needing external transformation scripts.
- **Dependency Pinning:** Exact dependency specifications are declared in requirements.txt and environment.yml, pinning compatible versions across Python 3.9 runtime, Scikit-learn, FastAPI, and MLflow.
- **MLflow Model Flavor:** Models were logged with mlflow.sklearn.log_model(), including Conda environment specifications and metadata tagging (Student Name: Ram Narayan, Roll: 2025af05183).

9. CI/CD Pipeline & Automated Testing Suite

A multi-stage Continuous Integration and Continuous Delivery (CI/CD) workflow was engineered in .github/workflows/mlops-pipeline.yml to enforce code quality, run unit tests, retrain models, and verify containers.



Automated Test Suite (14 Tests, 93% Code Coverage)

Test Module	Target Scope	Passed Tests	Coverage
tests/test_data_processing.py	Data integrity, cleaning, ColumnTransformer, no leakage	4 Passed	94%
tests/test_model.py	Model loading, output schema, accuracy benchmark (>80%)	4 Passed	95%
tests/test_api.py	Endpoints (/health, /predict, /predict/batch, /metrics)	6 Passed	86%
Total Pipeline Suite	End-to-end unit, integration, and performance validation	14 Passed, 0 Failed	93% Overall

CI/CD Automated Gate: *The pipeline automatically halts if any test fails, if Flake8 detects syntax anomalies, or if retraining accuracy falls below the 80.0% benchmark threshold.*

10. Containerization, Kubernetes & Observability

Production deployment was implemented using multi-stage Docker containerization, Kubernetes declarative manifests, Helm charts, and real-time Prometheus and Grafana monitoring.



Production Verification & Operations Runbook

- **Docker Hardening:** Multi-stage build (python:3.9-slim) runs as unprivileged user appuser (uid 1001), reducing container footprint and neutralizing privilege escalation vectors. Built-in HEALTHCHECK monitors /health.
- **Kubernetes Cluster Deployment:** Orchestrated in namespace mlops-heart-disease with 2 replicas, RollingUpdates, LoadBalancer Service on port 8000, NGINX Ingress, and Horizontal Pod Autoscaler (HPA 2-5 pods @ 70% CPU target).
- **Prometheus & Grafana Observability:** Custom metrics (api_requests_total, api_request_duration_seconds, model_predictions_total, model_prediction_confidence) are scraped every 5s to detect throughput, latency, and data drift in real time.

• ***Submission Verification:*** *All deliverables (code, tests, Helm charts, Dockerfiles, figures, 10-page report, and video script) are fully validated and packaged under Student ID:* ***2025af05183 (Ram Narayan)****.*